To Combine or not to Combine Graybox Crossover and Local Search?

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ABSTRACT

Specialized graybox local search and crossover have been successfully combined within the framework of the so-called DRILS (Deterministic recombination and iterated local search) algorithm. As for any evolutionary algorithm, the initial design framework, and the underlying high-level choices and parameters, are crucially important. The DRILS algorithm is no exception, and recent enhanced variants exist in the literature. In this paper, we aim at: (i) improving the performance of the latest variants of DRILS, and (ii) providing a better principled understanding of graybox search behavior and dynamics. On the basis of a preliminary analysis using Local Optima Networks of small-size NKQ-landscapes, we first highlight the difference of using local search with and without crossover. We then propose to pipeline these two techniques in a simple two-phase like iterated local search scheme which is shown to provide substantial improvements over the latest DRILS+ variant for large-size NKQ-landscapes. We further report a dedicated analysis in an attempt to provide new insights into the impact of local search and crossover on the phenotype and the genotype of the local optima encountered in the search trajectory.

CCS CONCEPTS

• **Theory of computation** → *Optimization with randomized search heuristics*; *Discrete optimization.*

KEYWORDS

Graybox optimization, Fitness landscape analysis

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1 INTRODUCTION

Context and goal. The evolutionary optimization community has made a steady progress in the design of efficient graybox search operators for tackling challenging optimization problems of increasingly large size [7, 25]. The specificity of graybox search operators is to take benefits from the knowledge available initially about the considered problem, at the aim of speeding up the search and helping convergence towards better high-quality solutions. Different specialized graybox operators can be found with respect to different optimization problems. These operators never come alone, and their effectiveness is tightly related to their smart integration in the framework of a specific evolutionary algorithm. Designing an evolutionary optimization process is however known to be a complex task. On the one hand, the high-level nature of evolutionary, and more generally stochastic, search algorithms implies a rich and large design space, which makes them difficult to engineer and to design efficiently. There is no general rule-of-thumb, but general design guidelines acknowledged by the community as crucially important. For example, a good balance between the intensification and diversification components is mandatory, the smart configuration of the underlying parameters is critical, etc. On the other hand, the accuracy of the design choices depends on the considered problem instance, and its landscape characteristics. This raises a number of additional research issues. For example, developing general-purpose (problem- and algorithm- independent) informative tools to characterize combinatorial landscapes is extremely difficult. Explaining the dynamics of a particular search process in light of the landscape properties is an active research topic, etc. Fitness-landscape analysis [14, 18, 26] is actually believed to provide useful techniques, to better understand the design issues one can face. The goal of this paper is to investigate new design choices when dealing with graybox optimization problems, while contributing to a better fundamental understanding of what makes such choices effective.

Related work and motivation. We consider the class of k-bounded pseudo-boolean functions [25], represented by the well-established NK-landscapes. NK-landscapes [10, 11] are defined as a linear combination of n sub-functions; each one accepting at most $k = K + 1$ variables, with K a parameter. The higher the value of K, the higher the degree of non-linearity among variables, and the more the underlying landscape is rugged and difficult to search. In this context, two main graybox techniques were developed by exploiting

the so-called *variable interaction graph (VIG)* [25], as available initially from the underlying sub-functions definition. The first technique is based on the efficient identification of improving bit-flip moves; hence, leading to a time-efficient local search hill-climber. The second technique is based on the gene transmission principle; hence, leading to state-of-the-art deterministic recombination operators [3, 4, 6, 22] allowing to efficiently find the best offspring among an eventually huge number of recombination possibilities.

In [6], it is shown how to combine these two techniques within the framework of a *Deterministic Recombination and Iterated Local Search (DRILS)* evolutionary algorithm. Besides, an alternative design framework was proposed there-in, and a comparison including to the so-called Gray-Box Parameterless Population Pyramid algorithm (GB-P3) [7], was conducted. DRILS was found to provide consistently better results, and it was adopted as a state-of-the-art design framework in subsequent investigations. DRILS is based on a simple design following the scheme of an *iterated local search* (ILS) algorithm [8, 13, 19]. More precisely, the graybox local search is used to find a first locally optimum parent, from which a second parent is computed by performing a perturbation followed by local search. An offspring is computed from the two parents by a graybox crossover, and the local search is again applied starting from the offspring. An acceptance criteria is applied to decide which solution is selected as the next first parent, and so on.

From a high-level perspective, DRILS has the following advantages: (i) the graybox local search is extremely fast in reaching locally optimum parents, (ii) the graybox crossover allows one to tunnel among local optima, and (iii) the general ILS scheme (i.e., perturbation and acceptance criteria) guides the whole search process towards promising unexplored regions. Interestingly, very recent advances have shown that, although the ILS scheme has a blackbox nature, it has a substantial impact on DRILS search performance. In [21], it is shown that designing graybox perturbation can help finding better solutions. The focus there-in was on learning variable linkage for blackbox problems of relatively small size, and no comparison with DRILS was included. Actually, only a standard ILS scheme was used for comparison, without including graybox recombination. In [1], the design of DRILS is revisited at the light of the perturbation and the acceptance criteria. New design options are investigated leading to substantially better variants. The intuition behind those variants is to design perturbation so as to complement the exploration power of graybox recombination. In this paper, we dig more deeply in the design of DRILS and its guiding principles, while having a two-fold goal: (i) improving the performance of the latest variants of DRILS, and (ii) providing a better principled understanding of graybox search behavior and dynamics.

Contribution overview. In order to leverage the design of DRILS, we first study the impact of the embedded graybox operators on small-size landscapes for which the search space can be fully enumerated. We then propose a new simple design for DRILS, as well as, a more focused analysis of the search dynamics for large-size landscapes. In more details, our contributions are as follows:

- In a preliminary step, we consider small-size NK-landscapes. We use Local Optima Networks (LONs) [17] to analyze the landscape global structure. We provide evidence that the LON edge density depends on whether crossover is used or not after local search; hence, suggesting that pure local search could expose some benefits over additionally activating crossover.
- In a second step, we leverage the design of DRILS for large-size NK-landscapes, by pipelining local search and crossover in a simple two-phase like scheme called PRILS (Pipelined recombination and iterated local search). Our experimental results, using NK-landscapes of up to one-million variables, show that PRILS performs substantially better compared to the latest DRILS variant.
- In a third step, we conduct a final analysis in an attempt to analyse more deeply the impact of activating or not activating the graybox crossover. Both a phenotype- and a genotype- oriented analysis of local optima as encountered during the search trajectory are reported. This analysis is mainly intended to provide a better understanding of the graybox search dynamics, hopefully paving the way for the developments of future more powerful techniques.

Outline. In Section 2, we recall some background. In Section 3, we report our preliminary analysis for small-size instances. In Section 4, we describe the PRILS algorithm, and report its performance on large-size instances. In Section 5, we conduct a final analysis on the graybox search behavior. In Section 6, we conclude the paper.

2 BACKGROUND

2.1 k-bounded functions and NKQ-landscapes

Let us consider n binary variables and an unconstrained pseudo-boolean function f (to be maximized) $f : \{0,1\}^n \mapsto \mathbb{R}$. The function is said to be k-bounded [25] if it can be decomposed as follows:

$$f(x) = \sum_{\ell=1}^{m} f^{(\ell)}(x_{i_{\ell,1}}, x_{i_{\ell,2}}, \ldots, x_{i_{\ell,j\leq k}})$$

where $i_{\ell,j}$ is the index of the jth variable ($j \leq k$) of the ℓth sub-function $f^{(\ell)}$. In a graybox setting, one have access to the previous decomposition and can supply the $f^{(\ell)}$ sub-functions. Following previous studies [3–5, 12], we consider NKQ (Quantized NK) landscapes as a representative class of k-bounded functions. These are defined by considering as many sub-functions as variables ($n = m = N$), and for every $\ell \in \{1, \ldots, n\}$, the sub-function $f^{(i)}$ is defined to depend on variable x_ℓ and $K = k - 1$ other variables. An additional positive integer parameter is used to define the codomain of each sub-function to the set $\{0, 1, \cdots, Q-1\}$.

Following previous work, we set $Q = 64$ and we consider a random model [9, 15] where the K epistatic interactions are selected at random, and the sub-functions values are generated at random.

2.2 DRILS and its variants

DRILS follows the framework of a hybrid iterated local search scheme combining efficient hill-climber and deterministic recombination [6]. This is recalled in the following, together with the latest DRILS variant, namely, DRILS+, as initially described in [1].

2.2.1 DRILS basic building blocks. Given a k-bounded function, a Variable Interaction Graph (VIG) [25] can be easily constructed to capture the non-linear interaction between variables. This is

Algorithm 1: A high-level pseudo-code of DRILS+ [1]

Parameters: Perturbation strength: $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$; time budget: $\boldsymbol{\tau}$

1 $x \leftarrow$ initial (random) solution;
2 $x \leftarrow$ GRAYBOX-LS(x);
3 **while** CPU-time $< \boldsymbol{\tau}$ **do**
4 $\quad y \leftarrow$ PERTURBATION($x, \boldsymbol{\alpha}$);
5 $\quad y \leftarrow$ GRAYBOX-LS(y);
6 $\quad z \leftarrow$ GRAYBOX-PX(x, y);
7 $\quad$ **if** $((x = z)$ || $(y = z))$ **then**
$\quad\quad$ // Original Drils design [6]: $x \leftarrow y$
$\quad\quad$ // Modification introduced in [1]
8 $\quad\quad y \leftarrow$ PERTURBATION($x, \boldsymbol{\beta}$);
9 $\quad\quad x \leftarrow$ GRAYBOX-LS(y);
10 $\quad$ **else**
11 $\quad\quad x \leftarrow$ GRAYBOX-LS(z);

either by assuming that every pair of variables appearing in the same sub-function have a nonlinear interaction, or by applying a Fourier transform [20] and then looking at the non-zero coefficients associated to a term with every two variables. Given the VIG, the change in fitness after a few number of variables are flipped can be computed more efficiently than when a full evaluation is performed. This is because a few number of sub-functions are likely to be affected. Based on this simple observation, and by efficiently tracking the (partial) fitness contributions of variables, an improving neighbor in the r-Hamming ball of a given solution can be computed in an extremely efficient manner [2, 5, 24]. In fact, assuming that each variable appears in $O(c)$ sub-functions, this requires $O((3kc)^r)$ time complexity (see Theorem 1 in [5]). Following the existing literature [1, 3–5], a first building block of DRILS is the local search, denoted GRAYBOX-LS, as obtained by a time-efficient first-ascent hill-climber at hamming distance 1. In the reminder, a local optima in our terminology is then with respect to this distance.

Deterministic recombination can also be designed in an effective manner. In this study, we will work with *Partition crossover* (PX) [22, 23]. By removing the VIG nodes corresponding to variables having the same assignment in two given parents, one ends up with a partitioning of the VIG into q connected sub-graphs, constituting the recombination components. The variables that appear together in the same component can then be inherited from one of the two parents; hence, leading to 2^q possible offspring. By observing that the VIG partitioning infers a linearly separable function, the best possible offspring can be computed in linear time in q. Partition crossover does exactly this; hence, being extremely efficient to find improving solutions in a huge proportion of the search space.

2.2.2 DRILS+. In DRILS, a locally optimum (first) parent is maintained at each iteration. A second parent is obtained by performing perturbation of strength α followed by graybox local search. The employed perturbation consists in flipping a proportion α of variables chosen randomly. The two parents are then recombined, to generate a new locally optimum parent, and the whole process is repeated by selecting one parent for the next iteration. This corresponds to a standard iterated local search (ILS) hybridized with deterministic recombination (PX). In [1], the design of DRILS is enhanced by revisiting the perturbation and the acceptance criteria components. Two new variants were considered and shown to significantly improve DRILS performance. In this paper, we consider the DRILS+ variant which was found the best performing one (see Table 1 in [1]). As depicted in Algorithm 1, DRILS+ has two perturbation parameters α and β, serving respectively to generate a second parent for recombination (line 4), and to escape to a new first parent, when the crossover is not successful (line 8). It is important to comment that the pseudo-code of Algorithm 1 hides a number of important technical details related to the efficient implementation of the embedded graybox operators. This includes the efficient retrieval of an improving neighbor when performing local search, the efficient computation of the best improving offspring for recombination, the efficient update of the so-called variable scores (or fitness contributions), and the management of the perturbation so as to optimize the update of the data structures used by the graybox operators at each iteration. Since the termination condition is set to a total amount of CPU-time τ, such considerations can have an impact on performance if not managed carefully. We use C as programming language and we took a special care having an extremely efficient implementation so as to have a fair assessment of the relative performance of the considered algorithms.

3 A LON-ORIENTED ANALYSIS OF SMALL-SIZE LANDSCAPES

In this section, we conduct a preliminary analysis focusing on small-size NKQ-landscapes, by using Local Optima Networks (LONs) [16, 17] as a tool to study the role of local search and recombination in DRILS. Generally speaking, LONs were introduced in the realm of combinatorial optimization and discrete search spaces, as a model to compress the information given by the whole search space into a smaller object, namely a graph, where the vertices are the local optima (LO) and the edges are the possible (weighted) transitions between them. As such, a LON allows one to visualize and to study the structure underlying a problem instance.

3.1 LONs with and without crossover

To compute the LON vertices, we adopt a full enumeration methodology to exhaustively compute the set of LOs, which can be done in a reasonable time for small-size landscapes. As for the LON weighted edges, they are usually meant to model how likely it is to jump or to escape from one LO to another one, typically when using an evolutionary operation; which can also be viewed as a kind of pair-wise distance between LOs. This can be defined on-purpose in different manners, so as to capture different topological features of the underlying search spaces. In our work, we propose to define the edges in two different ways respectively to the local search and the graybox crossover. Hence, we obtain two different LON variants for the same problem instance.

In the first LON variant, a *directed* edge $e_{i,j}$ exists between LO_i and LO_j, iff, starting from LO_i, there exists a non-zero probability $p_{i,j}$ to reach LO_j by performing a perturbation of strength α followed by a local search (i.e., exactly as if lines 4 and 5 of Algorithm 1 were executed in a sequence with $x := \text{LO}_i$). The second considered LON extends the first one by adding crossover in the edge

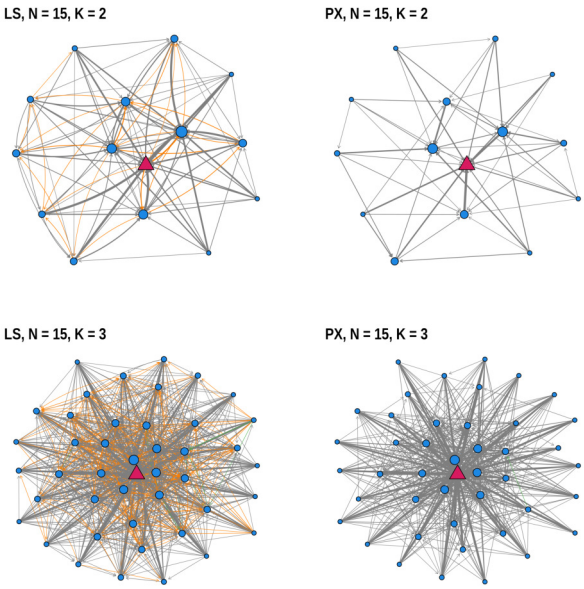


Figure 1: LON examples using local search (Right) and Local search with crossover (Left). Notice the N and K labels. The global optima is the red triangle. The size of LOs is proportional to their fitness. Orange edges indicate non-improving edges. Edge thickness is proportional to weights.

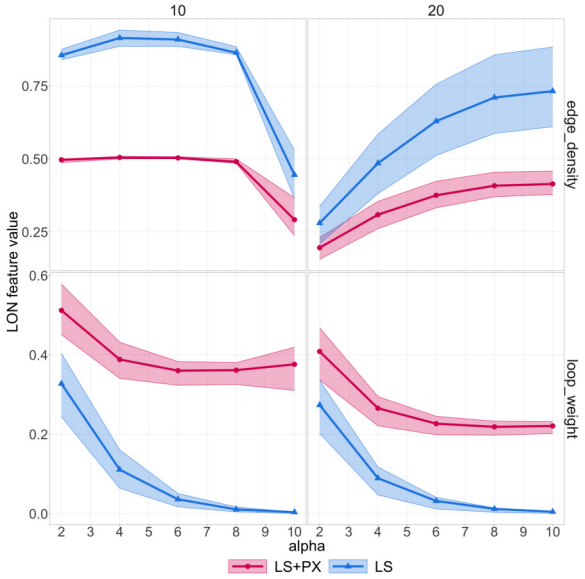


Figure 2: LON edge_density and loop_weight features as a function of α. From left to right, $N = 10$ and 20.

definition. More precisely, a *directed* edge $e_{i,j}$ exists iff LO_j can be reached from LO_i with a non-zero probability $p'_{i,j}$, by considering LO_i as a first parent, then performing perturbation and local search to find a second parent that is used for recombination to obtain an offspring, which is further improved by local search (i.e., exactly as if lines 4, 5, 6, and 11 of Algorithm 1 were executed in a sequence with $x := \mathrm{LO}_i$). The weight $w_{i,j}$ of an edge $e_{i,j}$ is simply defined respectively as $p_{i,j}$ and $p'_{i,j}$ for the two LON variants. Since these probabilities are difficult to compute, we estimate them by sampling 1000 times, starting from every possible LO and performing perturbation, local search, and crossover, i.e., the frequency of finding LO_j when sampling accordingly from LO_i is used as an estimate.

3.2 LON visual inspection and analysis

In Fig. 1, we show a few representative LON examples of NKQ-landscapes with $N = 15$, $K \in \{2, 3\}$, and $\alpha = 6$. For a better visualization, we only show the edges with weights falling in the 50 quantile, and we remove self-loops. We can see that the number of edges increases as the value of K, that is the landscape ruggedness, increases. More importantly, we can see that the number of edges is smaller in the second LON variant where crossover is used. This visual observation can be supported by a more rigorous analysis.

More specifically, we consider two simple graph features: (i) the edge density (edges_density), that is the number of edges relatively to the complete graph, and (ii) the average self-loop weight (loop_weight), that is the average weight of edges starting at a node and ending at the same node. For the purpose of our analysis, we generate 10 instances for each combination of $N \in \{10, 15, 20\}$ and $K \in \{2, 3, 4, 5\}$, i.e., 120 NKQ-landscapes, and construct the two LON variants for every perturbation strength $\alpha \in \{2, 4, 6, 8, 10\}$, i.e., 600 LONs. Due to space constraints, we only give an overview of feature values in Fig. 2, omitting $N = 15$ and aggregating over K.

From Fig. 2, we can clearly confirm that the LON density is significantly higher when *not* using crossover. Furthermore, the value of the perturbation α is found to have a high impact on the LON features. For example, for $N = 20$, higher α-values lead to more dense LONs with a smaller average self-loop weight. One should remark that the higher the weight of a self-loop, the higher the corresponding probability $p_{i,i}$ or $p'_{i,i}$, and the more likely to stay at the same local optima. In other words, the self-loop weight indicates how likely the underlying search procedure is able to escape from the current local optima. In this respect, we can see that although increasing α leads to smaller weight values overall, combining crossover and local search still leads to relatively high average self-loop weights, i.e., around 0.2 for the highest experimented α values. In contrast, applying local search alone makes the self-loop weight to consistently decrease until reaching 0, which means that it is very unlikely that the local search *without crossover* stays long (trapped) in a local optima.

On the one hand, even-though the graybox crossover is designed to explore efficiently a large number of possible offspring, our analysis suggests that it may imply less connectivity among LOs; which may eventually push the search trajectory towards specific regions that are difficult to escape. Besides, the perturbation may not be sufficient to help the crossover to avoid being trapped, which can interestingly explain the idea of using a second perturbation factor in the latest DRILS+ variant [1]. On the other hand, having larger densities, and smaller self-loop weights, can only indicate that it is eventually easier to visit more diversified local optima; but it does not necessarily mean that it is easier to reach a high-quality solution more quickly. Consequently, one can claim that pure local

Algorithm 2: PRILS (Pipelined Recombination and Iterated Local Search)

Parameters: Perturbation strength: $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\boldsymbol{\gamma}$; CPU time: $\boldsymbol{\tau}$; number of time slices: $\boldsymbol{\lambda}$

```
// Phase I: run Ils until 'convergence'
```

1 $\overline{f_{\text{prev}}}$, $\overline{f_{\text{curr}}} \leftarrow 0$; ▷ Average (slice) fitness;
2 $x \leftarrow$ initial (random) solution;
3 $x \leftarrow$ GRAYBOX-LS(x);
4 **while** `CPU-time-ils` $< \boldsymbol{\tau}$ **do**
  `// run Ils for one time slice` τ/λ
5 $\overline{f_{\text{curr}}} \leftarrow 0$; $i \leftarrow 0$;
6 **while** `CPU-time-slice` $< \boldsymbol{\tau}/\boldsymbol{\lambda}$ **do**
7 $y \leftarrow$ PERTURBATION($x, \boldsymbol{\gamma}$);
8 $y \leftarrow$ GRAYBOX-LS(y);
9 $x \leftarrow y$ ▷ Always accept local optima;
10 $\overline{f_{\text{curr}}} \leftarrow \overline{f_{\text{curr}}} + f(x)$; $i \leftarrow i + 1$;
11 **if** $\overline{f_{curr}}/i \geq \overline{f_{prev}}$ **then** $\overline{f_{\text{prev}}} \leftarrow \overline{f_{\text{curr}}}/i$;
12 **else break while**;

```
// Phase II: run Drils+ for the remaining time
```

13 **Run** DRILS+($x, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\tau}$ − `CPU-time-ils`) ;

search is more diversification-oriented compared to further using a graybox crossover. This claim can have strong consequences of how local search and crossover should be combined within an effective evolutionary optimization framework such as DRILS. One should of course mitigate this claim with the small-size setting we are considering, and the fact that the different other components and parameters of DRILS may enter into a complex interaction for optimal performance. In the next section, we shall nonetheless show that the latest DRILS+ variant can still be improved substantially based on such a simple claim.

4 COMBINING ILS AND DRILS FOR LARGE SIZE LANDSCAPES

In this section, we describe a new algorithm hybridizing local search and crossover, and study its performance for large-size instances.

4.1 Algorithm Description

4.1.1 Preliminaries. On the one hand, the graybox crossover is a powerful operator. On the other hand, using a pure local search algorithm has also some benefits, especially when the landscape is rugged. With respect to the initial design of DRILS, local search is mainly used to feed the crossover. Nonetheless, this was leveraged in the latest DRILS+ variant, where the perturbation and the local search are further used to avoid that the crossover gets trapped. In other words, consistently with our LON analysis, a side effect of using the powerful graybox crossover can be to guide the search in specific search regions that are difficult to escape. Thus, one can hypothesize that disabling the crossover could be beneficial for the search. This is exactly the rational behind the new algorithm described in this section.

4.1.2 PRILS := `exec` ILS | `exec` DRILS. Let us view ILS (without crossover) and DRILS (with crossover), as two high-level building-blocks that can be combined on purpose. We then investigate the idea of executing ILS and DRILS iteratively in a sequence. This idea can be set in several different manners. We consider a very simple two-phase scheme, where ILS is first executed, followed by DRILS; hence, ending up with PRILS (Pipelined recombination and iterated local search), as described in the template of Algorithm 2. The first phase of PRILS consists mainly in executing a standard ILS without crossover (outer 'while' loop in line 4, where `CPU-time-ils` encodes the global elapsed time). However, we manage to design an adaptive criteria to decide when to activate the second phase, and enable to further use crossover. In fact, we divide the overall available execution time τ into λ time slices, where λ is a user-defined parameter. At a given time slice (inner 'while' loop line 6, where `CPU-time-slice` encodes the elapsed time in the current slice), ILS is then executed while tracking the fitness of the different local optima. Notice that we use a non-elitist acceptance criteria where a newly generated local optima is always accepted regardless to its fitness. Notice also that fitness is never computed from scratch, since it can be computed incrementally while the embedded graybox local search is running. At the end of a current time slice, if the average fitness value of the so-encountered local optima is better than in the previous time slice (line 11), then the first phases simply continues executing ILS for a new time slice. Otherwise, the second phase is activated, which consists in running the latest DRILS+ variant for the remaining time budget, and using the last local optima (encountered in the first phase) as an initial solution.

4.1.3 Discussion. A number of observations can be made for the proposed design. Firstly, the ILS phase is motivated by giving the local search a chance to find interesting regions, which are further searched intensively using crossover. Of course, different design options could have been adopted, but as it will be shown in our experiments, this very simple two-phase design is surprisingly found to be extremely effective. Secondly, the PRILS algorithm has four main parameters, the perturbation strength α and β as appearing initially in the DRILS+ algorithm, the perturbation strength γ in the first ILS phase, and finally the number of time slices λ. The perturbation parameters are clearly related to the DRILS and ILS procedures. The λ parameter is inherently related to the design of PRILS. The λ parameter allows us to scale the duration of a time slice with respect to the overall considered time budget τ, which is different from a static (hard-coded) setting of the duration of a time slice. The duration of a time slice, as implied by the choice of λ, clearly impacts the current average fitness value of local optima. The rational behind our design is that tracking the overall quality of the local optima during a time slice provides an indication on the quality of the search region explored currently by ILS, which is seemingly different from tracking the evolution of the best so-far local optima. The condition allowing the activation of the second DRILS+ phase can then be roughly interpreted as the ILS has reached an interesting search region *relatively* to the previous slices. This can also be viewed as a convergence criteria, although the ILS may still have some ability in improving the best so-far local optima. As it will be shown in our experimental analysis, such a design is not only effective, but robust to the setting of the λ parameter.

4.2 Experimental Results

4.2.1 Experimental set-up. We consider large-size NKQ-landscapes with $N \in \{10^4, 3 \cdot 10^4, 10^5, 3 \cdot 10^5, 10^6\}$ and $K \in \{3, 5\}$. For each configuration, we generate 10 instances. We experiment both PRILS and DRILS+. We experiment two termination conditions $\tau \in \{1, 5\}$ minutes. For DRILS+, we set α and β in the range $\{0.01, 0.025, 0.05, 0.075, 0.1\}$. As for PRILS, in order to keep our experiments manageable, we restrict the parameter range as follows. We first extract the value of α and β providing the best median fitness, over all instances and runs of the same N and K, when executing DRILS alone for a given τ. We there-by fix α and β in PRILS to those extracted values. Then, we experiment PRILS with the two remaining parameters $\gamma \in \{0.0005, 0.001, 0.005, 0.01, 0.025, 0.05\}$ and $\lambda \in \{60, 90, 120\}$. Each algorithm configuration is executed for 20 independent runs on every instance. Overall, we have $5_{(N)} \times 2_{(K)} \times 10_{(\text{NKQ})} \times \left(\left(5_{(\alpha)} \times 5_{(\beta)}\right)_{\text{DRILS+}} + \left(5_{(\gamma)} \times 3_{(\lambda)}\right)_{\text{PRILS}}\right) \times 2_{(\tau)} \times 20_{(runs)} = 160000$ runs representing 333 CPU days. The experiments were conducted on a cluster equipped with `64 Intel Xeon Gold 6130 2.10` GHz processors. As mentioned before in Section 2.2.2, our algorithms are implemented in `C` while taking special care to have efficient processing of all graybox related operations. Finally, performance is given as the *relative fitness deviation* (rfd) of the best solution, say s, found by an algorithm configuration during a run, from the best solution, say $\tilde{s}^\star$, found by any algorithm run for a given instance, i.e., $100 \cdot (f(\tilde{s}^\star) - f(s))/f(\tilde{s}^\star)$. The lower the better.

4.2.2 Overall performance. In order to appreciate the relative performance of each algorithm when configured optimally, we compare PRILS and DRILS+ using the parameter values providing the best average rfd-value for a fixed τ, and for each group of instances of a given N and K. In Table 1, we summarize for each group the number of instances where PRILS performs statistically better, worst, or similarly, than DRILS+, using a Mann-Whitney statistical test with a p-value of 0.05. For $\tau = 5mn$ (respectively, $\tau = 1mn$) , PRILS is significantly better than DRILS+ on 85 (respectively, 86) over 100 considered NKQ-landscapes, while being never worst. Besides, the only instances where PRILS does not show significant improvement over DRILS are those with the smallest values of N or K. In Fig. 3, we further report the obtained rfd-values (averaged over instances and runs). We can clearly see that the gap between PRILS and DRILS+ increases substantially as a function of N and K. For the largest NKQ-landscapes, the difference between PRILS and DRILS+ is relatively impressive, which shows the superiority of the proposed two-phase PRILS design over the initial hybrid DRILS+ design.

4.2.3 Anytime profile. To better analyze the differences in performance, we show in Fig. 4 the convergence profile of both algorithms. In contrast to DRILS+ which has a standard convergence profile with a relatively smooth decrease in the rfd-values, the convergence profile of PRILS follows its underlying two-phase design. In a first phase, corresponding to ILS, PRILS shows a relatively slow progress compared to DRILS+. However, as soon as the second phase using crossover is activated, PRILS is able to not only catch up over DRILS+, but more surprisingly, to overtake DRILS+ very quickly, and to keep the lead for the reminder of the run. Notice in particular that DRILS+ is performing better than PRILS in the first phase, which is consistent with the fact that the ILS without crossover is worst than when

Table 1: Number of instances where PRILS is statistically better (▲), similar (≈), and worst (▽) compared to DRILS+.

	N	10000		30000		100000		300000		1000000	
τ	K	3	5	3	5	3	5	3	5	3	5
1 mn	▲	0	10	6	10	10	10	10	10	10	10
	≈	10	0	4	0	0	0	0	0	0	0
	▽	0	0	0	0	0	0	0	0	0	0
5 mn	▲	4	10	3	8	10	10	10	10	10	10
	≈	6	0	7	2	0	0	0	0	0	0
	▽	0	0	0	0	0	0	0	0	0	0

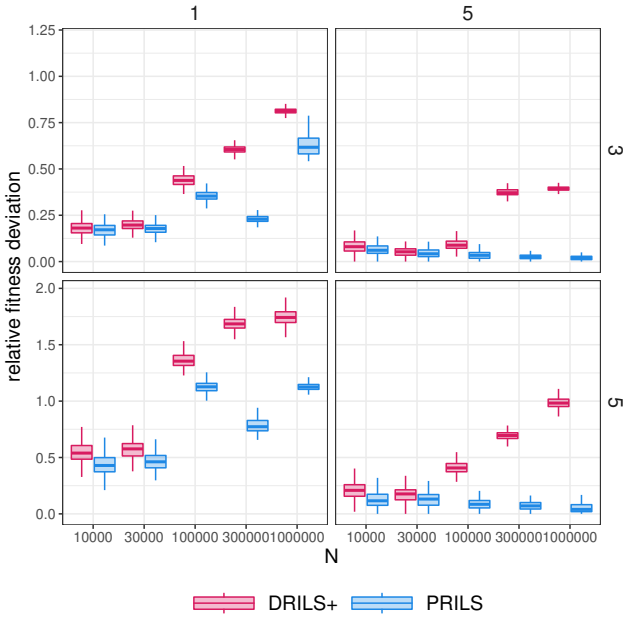


Figure 3: Rfd-values of PRILS and DRILS+ as a function of N. From left to right, CPU-time $\tau = 1$ and 5 minutes. From up to bottom, $K = 3$ and 5.

hybridized with crossover. However, the ability of PRILS to outperform DRILS+ during the second phase is clearly intriguing; given that the second phase consists in running DRILS+ (with exactly the same parameters) but starting from the last local optima found by ILS. Notice that this holds independently of the two experimented time budgets τ. In an attempt to better understand such a behavior, we shall provide in the next section a more focused analysis. For completeness, we first conclude our performance analysis in the next paragraph by commenting briefly on the parameter setting.

4.2.4 Parameter settings. In Table 2, we report the values of the parameters leading to the best median rfd-values. Remember that PRILS is configured with the best values of the (shared) perturbation parameters α and β obtained for DRILS+. The values of these two parameters are consistent with the initial work on DRILS+ (see Section 4.5 in [1]). With respect to the best perturbation strength γ underlying the first ILS phase of PRILS, we found that it is more beneficial for the search to have a smaller value than for the α and β

Table 2: Best parameters found for PRILS ($\tau = 5\ mn$), followed by the average rfd-values of PRILS with $\lambda \in \{60, 90, 120\}$, and the best average rfd-values obtained by DRILS+.

		Best parameter values				rfd with PRILS			rfd with DRILS+
N	K	α	β	γ	λ	$\lambda = 60$	$\lambda = 90$	$\lambda = 120$	
10000	3	0.075	0.025	0.0005	60	0.069	0.079	0.080	0.095
	5	0.025	0.025	0.01	120	0.161	0.158	0.150	0.243
30000	3	0.075	0.075	0.025	120	0.056	0.060	0.055	0.069
	5	0.025	0.05	0.01	90	0.197	0.193	0.201	0.233
100000	3	0.05	0.1	0.01	120	0.046	0.043	0.042	0.104
	5	0.025	0.05	0.01	60	0.066	0.093	0.093	0.386
300000	3	0.05	0.075	0.025	60	0.032	0.037	0.044	0.384
	5	0.01	0.05	0.01	60	0.061	0.086	0.100	0.692
1000000	3	0.05	0.05	0.025	60	0.015	0.020	0.032	0.386
	5	0.025	0.025	0.01	60	0.038	0.047	0.100	0.981

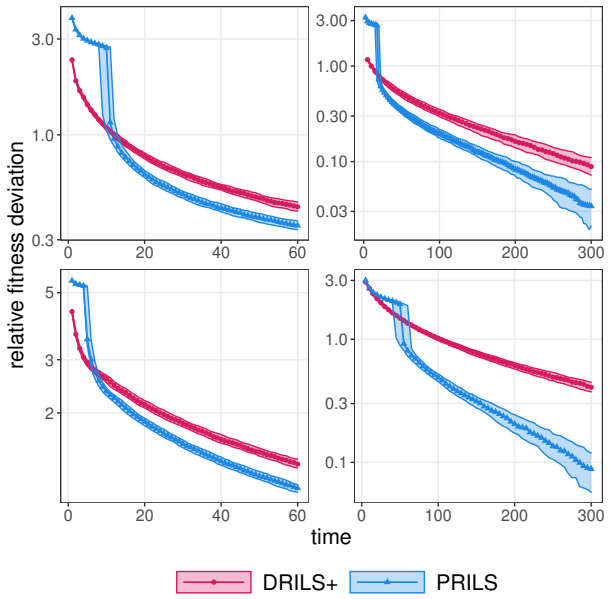


Figure 4: PRILS convergence profile $N = 10^5$. Left to right: CPU-time $\tau = 1$ and 5 minutes. Up to bottom, $K = 3$ and 5.

perturbations used in the second DRILS+ phase. More interestingly, we found that the choice of the number of time slices λ, which is inherent to the PRILS design, has only a limited impact on the relative performance. In fact, the rfd-values in the last four columns of Table 2 show that PRILS is robust to the value of λ, and can substantially outperform DRILS+ for all experimented λ-values.

5 A LO-ORIENTED ANALYSIS

Besides being able to beat DRILS+, the proposed design leads to a specific convergence profile of PRILS that we aim to better understand. Since PRILS was found to provide substantially better solutions in the second phase while using exactly the same search procedure than DRILS+, a reasonable hypothesis is that the local optima (LO) found by ILS during the first phase have some specific properties. To investigate more deeply such a hypothesis, we propose to analyse the set of LOs found respectively with and without crossover, as discussed in the following.

5.1 A phenotype oriented analysis

For the purpose of this first phenotype-oriented analysis, we conduct the following additional experiments. We consider a large-scale NKQ-landscape with $N = 10^5$ and $K = 5$. We generate 10 random instances and execute 10 independent runs of ILS and DRILS+, executed *separately* starting from the same randomly generated solution, and using the best perturbation values as given in Table 2, i.e., $\alpha = 0.025$ and $\beta = 0.05$ for DRILS+ and $\gamma = 0.01$ for ILS. For each execution, we record *all* the LO solutions encountered during the search trajectory. There-by, we collect *two sets* of LO solutions, denoted $\mathcal{LO}_{\text{ILS}}$ and $\mathcal{LO}_{\text{DRILS+}}$, by merging the LOs from the different (2×100) runs respectively to the two algorithms. *Independently for each of the two sampled LO-sets*, we iterate over every LO solution, say every x, and perform the regular local search and crossover steps, i.e., apply perturbation followed by local search to obtain a second parent, say y; then, recombine x and y by crossover followed by local search to obtain a LO x'. For each collected LO solution x, this is repeated 10 times using the *same* perturbation strength $\alpha = 0.025$ *invariantly* for two LO sets. We then compute the relative fitness improvement (rfi) of x' over x defined by $(f(x) - f(x'))/f(x)$; the higher the better. This statistic is reported in Fig. 5, as a function of the fitness level of the LO solutions x being respectively in $\mathcal{LO}_{\text{ILS}}$ and $\mathcal{LO}_{\text{DRILS+}}$.

Before analyzing the results in Fig. 5, it is important to correctly interpret the meaning of the rfi-values computed for *two* LO solutions *being at the same fitness level*, but belonging respectively to the two LO-sets. Computed for $\mathcal{LO}_{\text{DRILS+}}$, the rfi-values, at a given fitness level, can be considered as the empirical fitness improvement that can be expected by executing a regular iteration of DRILS+ (especially since we use the same α value when collecting the LO solutions and when computing the rfi-values). However, computed for $\mathcal{LO}_{\text{ILS}}$, the rfi-values render the (empirical) expected fitness improvement if *exactly the same* local search and crossover were executed for a *different* LO solution, *being at the same fitness level*, but found by iterated local search.

From Fig. 5, the first important observation is that the empirical improvement expected from applying exactly the same local search and crossover operators does not depend solely on the LO fitness level. In fact, the rfi-values are significantly different depending on whether the LO solutions are taken from $\mathcal{LO}_{\text{DRILS+}}$ or from $\mathcal{LO}_{\text{ILS}}$. This means that two LO solutions at the same fitness level, but with different genotypes, cannot be considered as two equally promising candidates for the underlying evolutionary graybox operators. The

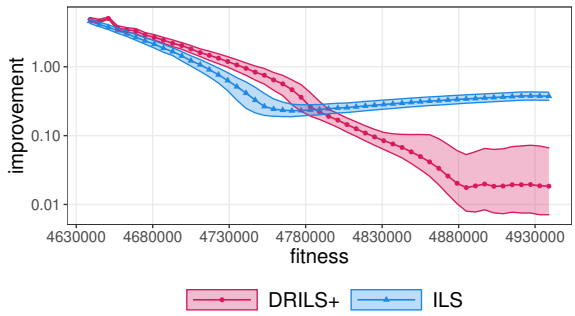


Figure 5: rfi-values (in ‰) as a function of LO fitness level.

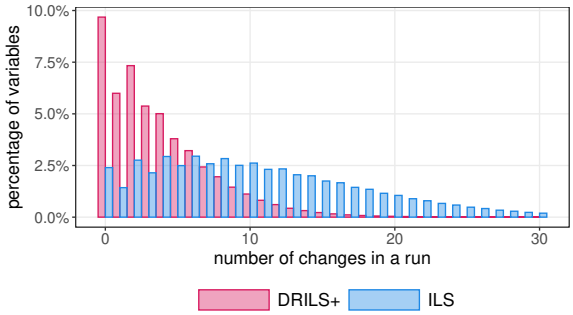


Figure 6: Histogram of the changes in LO variable assignment

second important observation is that the LO solutions visited by ILS can be surprisingly much more promising than the ones visited by DRILS+, which can actually explain the superiority of PRILS. In fact, the rfi-values computed with the LO solutions as visited by DRILS+ are better for low-quality fitness levels, whereas the opposite happens for higher fitness levels. This reminds the two-phase behavior observed when analyzing the convergence profile of PRILS. Combining these two observations, it appears that ILS plays the role of a warm-up procedure allowing PRILS to find genotype-promising LO solutions, from which DRILS+ (and the embedded graybox crossover) is able to escape more easily to high-quality basin of attractions. To reinforce the validity of such a claim, we provide a more focused genotype-oriented analysis.

5.2 A genotype-oriented analysis

Using the same NKQ-landscapes as in the previous section, we execute ILS (with $\alpha = 0.01$) and DRILS+ (with $\alpha = 0.025$ and $\beta = 0.05$) 100 independent runs for a fixed number of 5000 iterations, and starting from a randomly generated initial solution. For each run, we record the LO solutions considered at the beginning of each iteration during the search trajectory. For each binary variable, we compute the number of times its assigned value (0 or 1) in a LO solution has changed over the different iterations of a given run. Consequently, by aggregating over the different runs and instances, we are able to compute the percentage of variables whose assignment in the current LO solution changes for a given number of times during the execution of ILS or DRILS+, i.e., or alternatively the empirical probability that the assignment of a LO-variable changes for a given number of times during a run. This is reported in the histogram of Fig. 6.

We can see that ILS and DRILS imply very different distributions. Interestingly, DRILS+ has a tendency to change less frequently the variable assignment of visited LO solutions compared to ILS. For instance, 9.6% of the LO variables are never changed in DRILS+, whereas this drops to 2.3% for ILS. Only 3% of LO variables change more than 10 times during the 5000 iterations for DRILS+, which is to be contrasted to 22, 5% in the case of ILS. This indicates that DRILS+ is acting on a more restricted region of the search space compared to ILS, which is very likely because of the intensification power of the underlying graybox crossover. Consequently, although DRILS+ performs better than ILS, it is likely to intensify the search on specific region thanks to the graybox crossover, while lacking search diversification. In contrast, ILS is able to visit more diversified regions as captured by the diversity of the underlying LO genotypes, while lacking intensification in those visited regions. The proposed PRILS algorithm appears to take the benefits from these two aspects, although its design follows a very simple two-phase scheme.

6 CONCLUSION AND PERSPECTIVES

In this paper, we investigated the question of whether combining, or not combining, graybox crossover with local search, can impact search performance. We proposed a simple two-phase scheme which was shown to be extremely effective. Besides, we paid a special attention to provide a better understanding of search behavior and dynamics, by conducting an advanced fitness landscape analysis on small-size and large size instances. We believe that the knowledge gained from such an analysis is of fundamental interest for future research investigations. In particular, the preliminary LON analysis for small-size instances is to be leveraged by designing more advanced network features, hopefully allowing to better elicit the landscape structure as might be viewed by a target graybox search operator. More tools are also needed for analyzing the more challenging large-size setting, in light of the considered algorithmic components. The phenotype/genotype LO analysis provided in this paper shows that hidden information can be extracted to inform about the behavior of a given algorithm. We believe that such an information can be extremely useful to design improved algorithms; for example, by adapting the graybox components and parameters as the search process is running, and by inferring an enhanced balance between, the ability of graybox crossover to intensively search for an improving offspring, and the ability of the local search and perturbation to visit promising local optima.

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